

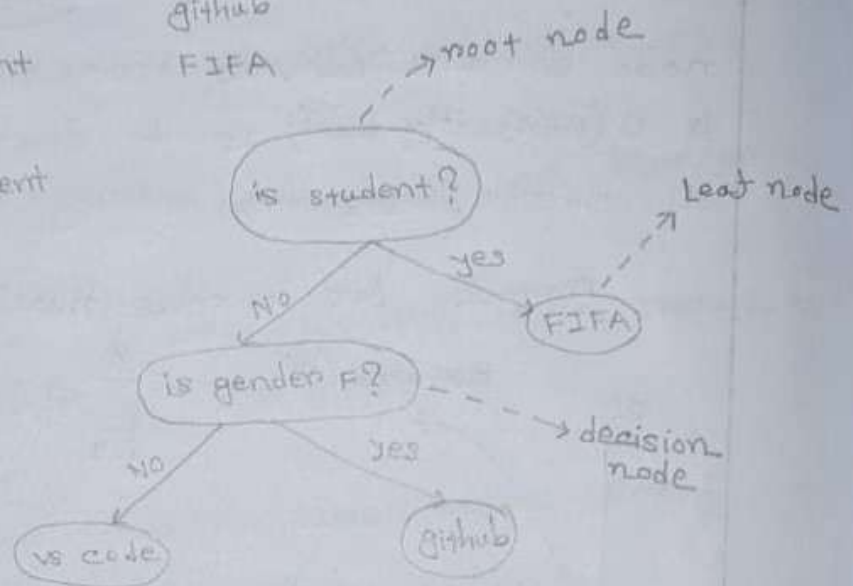
"Decision Tree"

gender	occupation	suggestion
F	student	FIFA
F	coder	github
M	coder	vs code
F	"	github
M	student	FIFA

if occupation == student
→ FIFA

else
if gender == F
→ github

else
→ vs code



Pseudo code

- ① Begin with a dataset with some col and classification or regression output.
- ② Find the best feature and ask question based on the ans. go to the next best feature
- ③ Do this till we get a leaf/target node.

Pure split: If at one node we only have one specific class count and all the other class count are zero then it's called pure split.

And this node is called leaf node.

How to check a split is pure or not?

2 ways → Entropy
→ Gini-Impurity

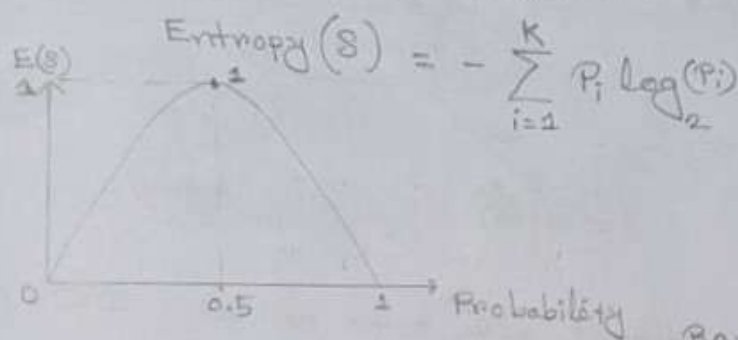
refers to how mixed or impure a set of class is. all same $\rightarrow$ 0 disorder
all equal $\rightarrow$ max n

How the features are selected?
Based on information gain.

Entropy (used in ID3 algorithm)

measure the uncertainty or disorder in a dataset. If a node contains samples from only one class, the entropy is 0 (perfectly pure). If it contains equal proportions of all classes, then the entropy is maximum (1).

Formula for a classification problem with K classes.



P_i = Proportion of class i samples in the node.

$\log_2$ is used to keep entropy in bits.

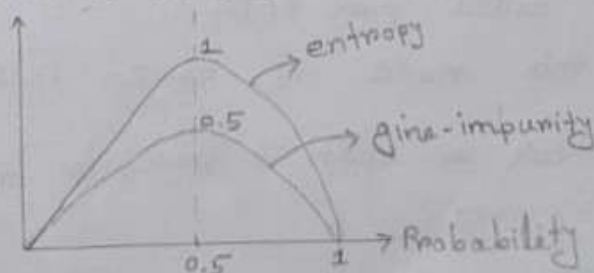
Range: 0 to 1

Gini Impurity (used in CART algorithm)

measures the probability of incorrectly classifying a randomly chosen element if it was randomly labeled according to the distribution of labels in the node.

$$\text{Gini}(S) = 1 - \sum_{i=1}^K P_i^2 \quad \left| \begin{array}{l} P_i \rightarrow \text{Proportion of class } i \\ \text{samples in the node} \end{array} \right.$$

range of gini: 0 to 0.5



It at a node: If we label the class based this
gini impurity is 0.9 node for a random sample then
the chances is 90% that it's wrong labeled.

Information Gain (IG)

a metric used in Decision Trees to select the best feature
to split the dataset. It measures the reduction in impurity
on uncertainty after a dataset is split based on a feature.

Intuition

→ when we split a dataset using a feature, so
"disorder" is removed.

→ Information gain tells how much of the disorder
was removed.

we want to maximize IG when choosing which feature to
split on.

Formula using Entropy:

Let, S be the parent node (the dataset before splitting)

A be a feature with values $v_1, v_2, v_3, \dots, v_n$

S_v is the subset of data where
feature $A = v$

IG of feature A is

$$IG(S, A) = Entropy(S) - \sum_{v \in \text{values}(A)} \frac{|S_v|}{|S|} \cdot Entropy(S_v)$$

Note

Higher IG = better feature to split

What is Pruning:

It's a process of removing sections of a decision tree that provide little to no predictive power, to reduce overfitting and improve generalization to unseen data.

→ Why pruning needed?

Decision Tree can easily overfit, especially when

- They grow too deep
- They perfectly classify the train data (low bias, high variance)
- Also they learn noise or outliers in the data

Benefits of Pruning

- Simpler and more interpretable trees.
- Better generalization.
- Less risk of fitting noise.

① Post Pruning (Cost complexity Pruning)

It's a technique used after growing a full decision tree. Where we remove branches or sub-trees that do not contribute significantly to the predictive power, to improve generalization and reduce overfitting.

Why post pruning?

→ post pruning aims to simplify the tree after it has perfectly fit to the train data without sacrificing much accuracy.

Cost complexity Pruning or weakest Link Pruning is the most widely used post pruning.

to balance

here define a cost function $\wedge$ the tree size and accuracy.

$$R_{\alpha}(T) = R(T) + \alpha \cdot |T|$$

where

T : a sub-tree

$R(T)$: Total misclassification error (impurity) of the tree on the training data.

$|T|$: num of leaf nodes in T subtree.

Pre-Pruning (Early Stopping)

a technique where the decision tree stops growing during the training phase - before it reaches full complexity.

Pre-Pruning Aims:

- Limit Complexity
- Improve Generalization
- Increase Training Speed.

How works?

The algorithm evaluates whether a node should be split based on one or more conditions:

- If condition is met $\rightarrow$ split the node
- If not $\rightarrow$ make it a leaf node and stop further splitting.

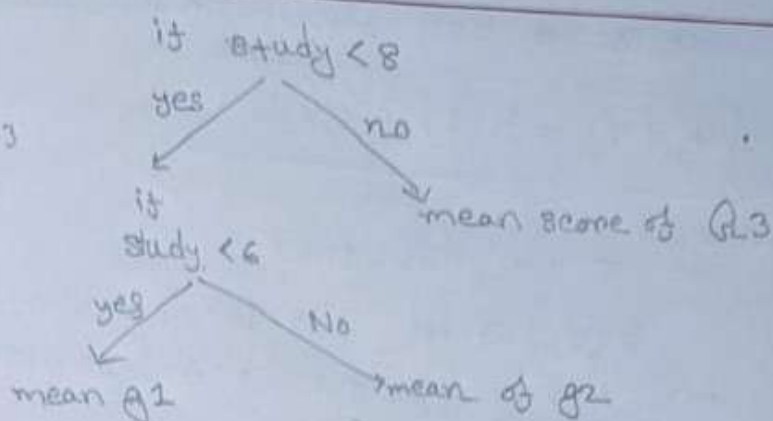
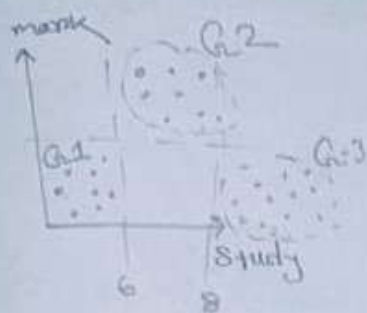
work How:

check

Tree depth $\geq$ max_depth
Node samples $<$ min_samples_split
Expected gain $<$ min_impurity_decrease
Num of leaf nodes $\geq$ max_leaf_nodes

} If any of these are true
then stop
otherwise split

Decision Tree (Regression)



But how the number will be selected if not fully separated groups are available.

Dataset			target (continuous)
Exp	gap	Salary	
2	Y	90K	Exp < 2
2.5	Y	42	
3	no	52	
4	Y	60	
4.5	Y	56	

Exp < 2

To do split on continuous column

at each node, it tries to minimize the MSE. (Variance Reduction)

It's data is splitted in 2 parts. $\{R_1, R_2\}$

$$\text{MSE before split: } \text{Var}_{\text{before}} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{y})^2$$

MSE after splitting into 2 part:

$$\text{Var}_{\text{after}} = \frac{N_1}{N} \sum_{i \in R_1} (x_i - \bar{y}_1)^2 + \frac{N_2}{N} \sum_{i \in R_2} (x_i - \bar{y}_2)^2$$

$$\text{Variance reduction} = \text{Var}(\text{before}) - \text{Var}(\text{after})$$

Split where variance reduction is high

CART Algorithm (greedy algorithm)

- at each node it can make at most 2 child.
- known as growing trees.

CART cost function for classification

$$J(k, t_k) = \frac{n_{\text{left}}}{n} G_{\text{left}} + \frac{n_{\text{right}}}{n} G_{\text{right}}$$

Instability

- orthogonal decision boundaries (at each node split based on only one feature).
- Sensitive to variation in train data.